

Bose–Einstein and Fermi–Dirac distributions

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In this chapter, we are going to consider the way in which *quantum mechanics* changes the statistical properties of gases. The crucial ingredient is the concept of *identical particles*. The results of quantum mechanics show that there are two types of identical particle: **bosons** and **fermions**. Bosons can share quantum states, while fermions cannot share quantum states. Another way of stating this is to say that bosons are not subject to the **Pauli exclusion principle**, while fermions are. This difference in ability to share quantum states (arising from what we shall call *exchange symmetry*) has a profound effect on the statistical distribution of these particles over the energy states of the system. This distribution over energy states is called the **statistics** of these particles, and we will demonstrate the effect of exchange symmetry on statistics. However, it can also be shown that another difference between bosons and fermions is the type of spin angular momentum that they may possess. This is enshrined in the **spin-statistics theorem**, which we will not prove but which states that bosons have integer spin while fermions have half-integer spin.

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Example 29.1

- Examples of bosons include: photons (spin 1), ^4He atoms (spin 0).
- Examples of fermions include: electrons (spin $\frac{1}{2}$), neutrons (spin $\frac{1}{2}$), protons (spin $\frac{1}{2}$), ^3He atoms (spin $\frac{1}{2}$), ^7Li nuclei (spin $\frac{3}{2}$).

29.1 Exchange and symmetry

In this section, we will argue why a two-particle wave function can be either symmetric or antisymmetric under exchange of particles. Consider two identical particles, one at position $\mathbf{r}_1$ and the other at position $\mathbf{r}_2$. The wave function which describes this is $\psi(\mathbf{r}_1, \mathbf{r}_2)$. We now define an exchange operator $\hat{P}_{12}$ which exchanges particles 1 and 2. Thus

$$\hat{P}_{12}\psi(\mathbf{r}_1, \mathbf{r}_2) = \psi(\mathbf{r}_2, \mathbf{r}_1). \quad (29.1)$$

Since the particles are identical, we also expect that the Hamiltonian $\hat{\mathcal{H}}$ which describes this two-particle system must commute with $\hat{P}_{12}$, i.e.

$$[\hat{\mathcal{H}}, \hat{P}_{12}] = 0, \quad (29.2)$$

so that the energy eigenfunctions must be simultaneously eigenfunctions of the exchange operator. However, because the particles are identical, swapping them over must have no effect on the probability density. Thus

$$|\psi(\mathbf{r}_1, \mathbf{r}_2)|^2 = |\psi(\mathbf{r}_2, \mathbf{r}_1)|^2. \quad (29.3)$$

¹A Hermitian operator has real eigenvalues, so is useful for representing real physical quantities in quantum mechanics.

If $\hat{P}_{12}$ is a Hermitian¹ operator, it must have real eigenvalues, so we expect that $\hat{P}_{12}\psi = \lambda\psi$, where λ is a real eigenvalue. Equation 29.3 shows that the only solution to this is $\lambda = \pm 1$, i.e.

$$\hat{P}_{12}\psi(\mathbf{r}_1, \mathbf{r}_2) = \psi(\mathbf{r}_2, \mathbf{r}_1) = \pm\psi(\mathbf{r}_1, \mathbf{r}_2). \quad (29.4)$$

The wave function must therefore have one of two types of **exchange symmetry**, as follows:

- The wave function is **symmetric** under exchange of particles:

$$\psi(\mathbf{r}_2, \mathbf{r}_1) = \psi(\mathbf{r}_1, \mathbf{r}_2), \quad (29.5)$$

and the particles are called bosons.

- The wave function is **antisymmetric** under exchange of particles:

$$\psi(\mathbf{r}_2, \mathbf{r}_1) = -\psi(\mathbf{r}_1, \mathbf{r}_2), \quad (29.6)$$

and the particles are called fermions.

This argument is valid for particles in three dimensions, the situation we usually encounter in our three-dimensional world, but fails in two dimensions. This occurs because you have to be a bit more careful than we've been here about how you exchange two particles. This point is rather an esoteric one, but the interested reader can follow up this point in the box on page 327 and in the further reading.

29.2 Wave functions of identical particles

In the previous section, we wrote down a two-particle wave function $\psi(\mathbf{r}_2, \mathbf{r}_1)$ which labelled the particles according to their position. However, there are lots more ways in which one could label a particle, such as which orbital state it is in, or what its momentum is. To keep things completely general, we will label the particles according to their state in a more abstract way. The effect on the statistics will then be more transparent, and is demonstrated by the following example (on page 328).

Anyons

The argument that we have used to describe exchange symmetry is, in fact, only strictly valid in three dimensions. In two dimensions, there are further possibilities other than fermions and bosons. For the interested reader, we give a more detailed description in this box.

We begin by noticing that eqn 29.3 allows the solution $\psi(\mathbf{r}_2, \mathbf{r}_1) = e^{i\theta} \psi(\mathbf{r}_1, \mathbf{r}_2)$, where θ is a phase factor. Thus exchanging identical particles means that the wave function acquires a phase θ . Defining $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$, the action of exchanging the position coordinates of two particles involves letting this vector execute some path from $\mathbf{r}$ to $-\mathbf{r}$, but avoiding the origin so that the two particles do not ever occupy the same position.

We therefore can imagine the exchange of particles as a path in $\mathbf{r}$ -space. Without loss of generality, we can keep $|\mathbf{r}|$ fixed, so that in the process of exchanging the two particles, they move relative to each other at a fixed separation. Thus, for the case of three dimensions, the path is on the surface of a sphere in $\mathbf{r}$ -space. Since the two particles are identical, opposite points on the surface of the sphere are equivalent and must be identified (giving $\mathbf{r}$ -space the topology of, what is known as, real two-dimensional projective space). It turns out that all paths on this surface fall into two classes:

Fig. 29.1 Paths in $\mathbf{r}$ -space, for the three-dimensional case, corresponding to (a) no exchange of particles and (b) exchange of particles.

ones which are contractible to a point [and thus correspond to no exchange of particles, yielding $\theta = 0$ to ensure the wavefunction is single-valued; see Fig. 29.1(a)] and those which are not [and thus correspond to exchange of particles; see Fig. 29.1(b)]. For this latter case we have to assign $\theta = \pi$, so that two exchanges correspond to no exchange, i.e.

$e^{i\theta} e^{i\theta} = 1$, so that $\theta = \pi$. This argument thus justifies that the phase factor $e^{i\theta} = \pm 1$, giving rise to bosons ($e^{i\theta} = +1$) and fermions ($e^{i\theta} = -1$).

However, the argument fails in two dimensions. In the two-dimensional case, the path is on a circle in $\mathbf{r}$ -space in which opposite points on the circle are equivalent and are identified. In this case, the paths in $\mathbf{r}$ -space can wind round the origin an integer number of times. This means that two successive exchanges of the particles [as shown in Fig. 29.2(c)] are not topologically equivalent to zero exchanges [if performed by winding round the origin in the same direction, as shown in Fig. 29.2(a)] and thus the phase θ can take any value. (In this case, $\mathbf{r}$ -space has the topology of real one-dimensional projective space, which is the same as that of a circle.) The resulting particles have more complicated statistical properties than either bosons or fermions and are called **anyons** (because θ can take ‘any’ value). Since θ/π is no longer forced to be ± 1 , and can take any *fractional* value in between, anyons can have **fractional statistics**. The crucial distinction between $\mathbf{r}$ -space in two and three dimensions is that the removal of the origin in two-dimensional space makes the space multiply connected (allowing paths which wind around the origin), whereas three-dimensional space remains singly connected (and a path which tries to wind round the origin can be deformed into one which does not).

We live in a three-dimensional world, so is any of this relevant? In fact, anyons turn out to be important in the **fractional quantum Hall effect**, which occurs in certain two-dimensional electron systems under high magnetic field. For more details concerning anyons, see the further reading.

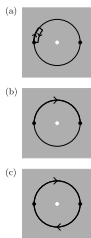


Fig. 29.2 Paths in $\mathbf{r}$ -space, for the two-dimensional case, corresponding to (a) no exchange, (b) a single exchange and (c) two exchanges of particles.

Example 29.2

Imagine that a particle can exist in one of two states, which we will label $|0\rangle$ and $|1\rangle$. We now consider two such particles, and describe their joint state by a product state. Thus

$$|0\rangle|1\rangle \quad (29.7)$$

describes the state in which the first particle is in state 0 and the second particle is in state 1. What are the possible states for this system if the particles are (a) distinguishable, (b) indistinguishable, but classical, (c) indistinguishable bosons, and (d) indistinguishable fermions?

Solution:

- (a) *Distinguishable particles:*

There are four possible states, which are

$$|0\rangle|0\rangle, \quad |1\rangle|0\rangle, \quad |0\rangle|1\rangle, \quad |1\rangle|1\rangle \quad (29.8)$$

- (b) *Indistinguishable, but classical, particles:*

There are now only three possible states, which are

$$|0\rangle|0\rangle, \quad |1\rangle|0\rangle, \quad |1\rangle|1\rangle \quad (29.9)$$

Since the particles are indistinguishable, there is no way of distinguishing the state $|1\rangle|0\rangle$ from $|0\rangle|1\rangle$.

- (c) *Indistinguishable bosons:*

There are also only three possible states. Clearly both $|0\rangle|0\rangle$ and $|1\rangle|1\rangle$ are eigenstates of the exchange operator, but $|1\rangle|0\rangle$ and $|0\rangle|1\rangle$ are not. However, if we make a linear combination²

$$\frac{1}{\sqrt{2}}(|1\rangle|0\rangle + |0\rangle|1\rangle), \quad (29.10)$$

this will be an eigenstate of the exchange operator with eigenvalue 1. Thus the three possible states are:

$$|0\rangle|0\rangle, \quad |1\rangle|1\rangle, \quad \frac{1}{\sqrt{2}}(|1\rangle|0\rangle + |0\rangle|1\rangle). \quad (29.11)$$

- (d) *Indistinguishable fermions:*

No two fermions can be in the same quantum state (by the Pauli exclusion principle), so $|0\rangle|0\rangle$ and $|1\rangle|1\rangle$ are not allowed. Thus only one state is possible, which is

$$\frac{1}{\sqrt{2}}(|1\rangle|0\rangle - |0\rangle|1\rangle). \quad (29.12)$$

This wave function is an eigenstate of the exchange operator with eigenvalue -1 .

²This example demonstrates quantum-mechanical **entanglement**. The states of the two particles are entangled, in the sense that if one particle is in the $|0\rangle$ state, the other particle has to be in the $|1\rangle$ state, and vice versa.

In general, for fermions, the requirement that $\hat{P}_{12}|\psi\rangle = -|\psi\rangle$ means that if $|\psi\rangle$ is a two-particle state consisting of two particles in the *same* quantum state, i.e. if $\psi = |\varphi\rangle|\varphi\rangle$, then

$$\hat{P}_{12}|\varphi\rangle|\varphi\rangle = |\varphi\rangle|\varphi\rangle = -|\varphi\rangle|\varphi\rangle, \quad (29.13)$$

so that

$$|\varphi\rangle|\varphi\rangle = 0, \quad (29.14)$$

i.e. the doubly-occupied state cannot exist. This, again, illustrates the Pauli exclusion principle, namely that two identical fermions cannot coexist in the same quantum state.

29.3 The statistics of identical particles

In the last section, we have demonstrated that exchange symmetry has an important effect on the statistics of two identical particles. Now we want to do the same for cases in which we have many more than two identical particles. Our derivation will be easiest if we do this by finding the grand partition function $\mathcal{Z}$ (see Section 22.3) for a system comprised either of fermions or bosons. In this approach, the total number of particles is not fixed, and this is an easy constraint to apply as we shall see. If one is treating a system in which the number of particles *is* fixed, we can always fix it at the end of our calculation. Our method will be to use the expression $\mathcal{Z} = \sum_{\alpha} e^{\beta(\mu N_{\alpha} - E_{\alpha})}$ (from eqn 22.20). Here, α denotes a particular state of the system. We assume that there are certain possible quantum states in which to place our particles, and that the energy cost of putting a particle into the i^{th} state is given by E_i . We will put n_i particles into the i^{th} state; here n_i is called the **occupation number** of the i^{th} state. A particular configuration of the system is then described by the product

$$\left[e^{\beta(\mu - E_1)} \right]^{n_1} \times \left[e^{\beta(\mu - E_2)} \right]^{n_2} \times \dots = \prod_i e^{n_i \beta(\mu - E_i)}. \quad (29.15)$$

The grand partition function is the sum of such products for all sets of occupation numbers which are allowed by the symmetry of the particles. Hence

$$\mathcal{Z} = \sum_{\{n_i\}} \prod_i e^{n_i \beta(\mu - E_i)}, \quad (29.16)$$

where the symbol $\{n_i\}$ denotes a set of occupation numbers allowed by the symmetry of the particles.

Fortunately, the total number of particles $\sum_i n_i$ does not have to be fixed,³ because that would have been a fiddly constraint to apply to this expression. In fact, we will only be considering two cases: fermions, for which $\{n_i\} = \{0, 1\}$ (independent of i), and bosons, for which $\{n_i\} = \{0, 1, 2, 3, \dots\}$ (independent of i). This allows us to factor out the terms in the product for each state i and hence write

$$\mathcal{Z} = \prod_i \sum_{\{n_i\}} e^{n_i \beta(\mu - E_i)}. \quad (29.17)$$

³The total number of particles is not fixed in the grand canonical ensemble, which is the one we are using here.

Example 29.3

Evaluate $\ln \mathcal{Z}$ for a gas of (i) fermions and (ii) bosons.

Solution:

(i) For fermions, each state can either be empty or singly occupied, so that $\{n_i\} = \{0, 1\}$, and hence eqn 29.17 becomes

$$\mathcal{Z} = \prod_i (1 + e^{\beta(\mu - E_i)}). \quad (29.18)$$

Hence

$$\ln \mathcal{Z} = \sum_i \ln(1 + e^{\beta(\mu - E_i)}). \quad (29.19)$$

(ii) For bosons, each state can contain any integer number of particles, so that $\{n_i\} = \{0, 1, 2, 3, \dots\}$, and hence eqn 29.17 becomes

$$\mathcal{Z} = \prod_i (1 + e^{\beta(\mu - E_i)} + e^{2\beta(\mu - E_i)} + \dots) \quad (29.20)$$

and therefore, by summing this geometric series, we have that

$$\mathcal{Z} = \prod_i \frac{1}{1 - e^{\beta(\mu - E_i)}}, \quad (29.21)$$

and hence

$$\ln \mathcal{Z} = - \sum_i \ln(1 - e^{\beta(\mu - E_i)}). \quad (29.22)$$

Summarizing the results of the previous example, we can write

$$\ln \mathcal{Z} = \pm \sum_i \ln(1 \pm e^{\beta(\mu - E_i)}), \quad (29.23)$$

where the $\pm$ sign means $+$ for fermions and $-$ for bosons.

The number of particles in each energy level is given by

$$\langle n_i \rangle = -\frac{1}{\beta} \left(\frac{\partial \ln \mathcal{Z}}{\partial E_i} \right) = \frac{e^{\beta(\mu - E_i)}}{1 \pm e^{\beta(\mu - E_i)}}, \quad (29.24)$$

and hence dividing top and bottom by $e^{\beta(\mu - E_i)}$ gives

$$\langle n_i \rangle = \frac{1}{e^{\beta(E_i - \mu)} \pm 1}, \quad (29.25)$$

where, again, the $\pm$ sign means $+$ for fermions and $-$ for bosons.

If μ and T are fixed for a particular system, eqn 29.25 shows that the mean occupation of the i^{th} state, $\langle n_i \rangle$, is a function only of the energy E_i . It is therefore convenient to consider the **distribution function** $f(E)$ for fermions and bosons, which is defined to be the mean occupation of

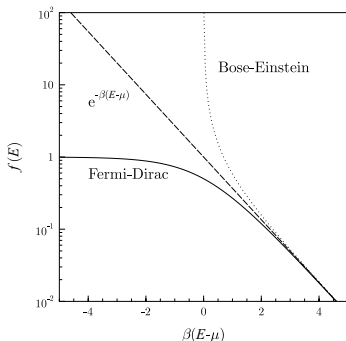


Fig. 29.3 The Fermi–Dirac and Bose–Einstein distribution functions.

a state with energy E . We can therefore immediately write down the distribution function $f(E)$ for fermions as

$$f(E) = \frac{1}{e^{\beta(E-\mu)} + 1}, \quad (29.26)$$

which is known as the **Fermi–Dirac distribution function**, and for bosons as

$$f(E) = \frac{1}{e^{\beta(E-\mu)} - 1}, \quad (29.27)$$

which is known as the **Bose–Einstein distribution function**. Sometimes the term on the right-hand side of eqn 29.26 is referred to as the **Fermi factor** and the term on the right-hand side of eqn 29.27 is referred to as the **Bose factor**. These are sketched in Fig. 29.3. Note that in the limit $\beta(E-\mu) \gg 1$, both functions tend to the Boltzmann distribution $e^{-\beta(E-\mu)}$. This is because this limit corresponds to low-density (μ small) and here there are many more states thermally accessible to the particles than there are particles; thus double occupancy never occurs and the requirements of exchange symmetry become irrelevant and both fermions and bosons behave like classical particles. The differences, however, are particularly felt at high density. In particular, note that the

distribution function for bosons diverges when $\mu = E$. Thus for bosons, the chemical potential must always be below, even if only slightly, the lowest-energy state. If it is not, then the lowest-energy state would become occupied with an infinite number of particles, which is unphysical. The implications for the properties of quantum gases will be considered in the next chapter.

Chapter summary

- The wave function of a pair of bosons is symmetric under exchange of particles, while the wave function of a pair of fermions is anti-symmetric under exchange of particles.
- Bosons can share quantum states, while fermions cannot share quantum states.
- Bosons obey Bose–Einstein statistics, given by

$$f(E) = \frac{1}{e^{\beta(\mu-E)} - 1},$$

while fermions obey Fermi–Dirac statistics, given by

$$f(E) = \frac{1}{e^{\beta(\mu-E)} + 1}.$$

Further reading

More information about anyons may be found in Canright and Girvin (1990), Rao (1992) and in the collection of articles in Shapere and Wilczek (1989).

Exercises

- (29.1) Differentiate between particles that obey Bose–Einstein and Fermi–Dirac statistics, giving two examples of each.

- (29.2) For the particles considered in Example 29.2, what is the probability that both particles are in the $|0\rangle$ state when (a) distinguishable, (b) indistinguishable, but classical, (c) indistinguishable bosons, and (d) indistinguishable fermions?

- (29.3) By rewriting the Fermi–Dirac function

$$f(E) = \frac{1}{e^{\beta(E-\mu)} + 1} \quad (29.28)$$

as

$$f(E) = \frac{1}{2} \left(1 - \tanh \frac{1}{2} \beta(E - \mu) \right), \quad (29.29)$$

show that $f(E)$ is symmetric about $E = \mu$ and sketch it. Find simplified expressions for $f(E)$

when (i) $E \ll \mu$, (ii) $E \gg \mu$ and (iii) E is very close to μ .

(29.4) Are identical particles always indistinguishable?

(29.5) Hydrogen (H_2) gas can exist in two forms. If the proton spins are in an exchange symmetric triplet ($S = 1$) state, it is known as **ortho-hydrogen**. If the proton spins are in an exchange antisymmetric singlet ($S = 0$) state, it is known as **para-hydrogen**. The symmetry of the total wave function must be antisymmetric overall, so that the rotational part of the wave function must be antisymmetric for orthohydrogen (so that the angular momentum quantum number J is 1, 3, 5, ...) or symmetric for parahydrogen (so that $J = 0, 2, 4, \dots$). The proton separation in hydrogen is 7.4×10^{-11} m. Estimate the spacing in Kelvin between the ground state and first excited state in parahydrogen.

Show that the ratio f of ortho-hydrogen to parahydrogen is given by

$$f = 3 \frac{\sum_{J=1,3,5,\dots} (2J+1) e^{-J(J+1)\hbar^2/2Ik_B T}}{\sum_{J=0,2,4,\dots} (2J+1) e^{-J(J+1)\hbar^2/2Ik_B T}}, \quad (29.30)$$

and find f at 50 K.

(29.6) In this exercise, we derive Fermi-Dirac and Bose-Einstein statistics using the microcanonical ensemble.

(a) Show that the number of ways of distributing n_j fermions among g_j states with not more than one particle in each is

$$\Omega_j = \frac{g_j!}{n_j(g_j - n_j)!}. \quad (29.31)$$

Here, j labels a particular group of states. Hence the entropy S is given by

$$S = k_B \ln \left[\prod_j \frac{g_j!}{n_j(g_j - n_j)!} \right]. \quad (29.32)$$

Hence show (using Stirling's approximation) that

$$S = -k_B \sum_j g_j [\bar{n}_j \ln \bar{n}_j + (1 - \bar{n}_j) \ln(1 - \bar{n}_j)], \quad (29.33)$$

where $\bar{n}_j = n_j/g_j$ are the mean occupation numbers of the quantum states. Maximize this expression subject to the constraint that the total energy E and number of particles N are constant, and hence show that

$$\bar{n}_j = \frac{1}{e^{\alpha + \beta E_j} + 1}. \quad (29.34)$$

(b) Show that the number of ways of distributing n_j bosons among g_j states with any number of particles in each is

$$\Omega_j = \frac{(g_j + n_j - 1)!}{n_j(g_j - n_j)!}. \quad (29.35)$$

Hence show that

$$S = k_B \sum_j g_j [(1 + \bar{n}_j) \ln(1 + \bar{n}_j) - \bar{n}_j \ln \bar{n}_j]. \quad (29.36)$$

Maximize this expression subject to the constraint that the total energy E and number of particles N are constant, and hence show that

$$\bar{n}_j = \frac{1}{e^{\alpha + \beta E_j} - 1}. \quad (29.37)$$